

Test 1

Diya Jain

1. This is a block cipher because the encryption operates on fixed-size blocks of data.

$$m = \underbrace{1010}_{m_1} \underbrace{1111}_{m_2} \underbrace{0101}_{m_3}$$

$$K_1 = 1110$$

$$K_2 = 0111$$

Apply P_1 on m_1

$$m_1 = 1010, \quad P_1(1010) = 0101 = m_1$$

Apply P_2 on m_3

$$m_3 = 0101, \quad P_2(0101) = 1010 = m_3$$

$$C_1 = m_1 \oplus K_1 = 0101 \oplus 1110 = 1011$$

$$C_2 = m_1 \oplus m_2 \oplus m_3 = 0101 \oplus 1111 \oplus 1010 = 0000$$

$$C_3 = m_3 \oplus K_2 = 1010 \oplus 0111 = 1101$$

$$C = 1011 \ 0000 \ 1101$$

$$m_1 = C_1 \oplus K_1 = 0010 \oplus 1110 = 1100$$

$$\text{Applying } P_1 \text{ to } m_1 = P(1100) = 0110$$

$$m_2 = m_1 \oplus C_2 \oplus m_3 = 1100 \oplus 0000 \oplus 1010 = 1111$$

$$m_3 = c_3 \oplus k_2 = 1101 \oplus 0111 = 1010$$

$$\text{Apply } P_2 \text{ to } m_3 = P_2(1010) = 0101$$

$$\therefore m = 0110 \ 1111 \ 0101$$

$$\bullet c_2 = m_1 \oplus m_2 \oplus m_3 = 1 \dots 1$$

$$\bullet c_2 = m_1 \oplus m_2 \oplus m_3 = 0 \dots 0$$

• The key size $2n$ is smaller than the message size $3n$, this scheme does not have perfect secrecy.

• In the semantic security game:

- $P(W_0) = 0$. Indeed, in Experiment 0, we always have $c_2 = 1 \dots 1$ so the adversary returns 1 with probability 1.

- $P(W_1) = 1$, Indeed, in Experiment 1, we always have $c_2 = 0 \dots 0$ so the adversary returns 0 with probability 1.

$$- \mathcal{A} = |P(W_0) - P(W_1)| = |0 - 1| = 1$$

$$2. \ K = \overset{1}{1} \overset{2}{1} \overset{3}{0} \overset{4}{0} \overset{5}{0} \overset{6}{0} \overset{7}{1} \overset{8}{1} \overset{9}{1} \overset{10}{1}$$

$$G(1) = k_1 = 1$$

$$G(4) = k_4 = 0$$

$$G(7) = k_7 = 1$$

$$G(2) = k_2 = 1$$

$$G(5) = k_5 = 0$$

$$G(8) = k_8 = 1$$

$$G(3) = k_3 = 0$$

$$G(6) = k_6 = 1$$

$$G(9) = k_9 = 1$$

$$G_{10} = k_{10} = 1$$

$$G_{11} = G_{10+1}(k) = 1 \oplus k_{(10 \bmod 10)+1} \oplus G_{10}(k)$$

$$= 1 \oplus 1 \oplus 1$$

$$= 1$$

$$G_{12} = G_{11+1}(k) = 1 \oplus k_{(11 \bmod 10)+1} \oplus G_{11}(k)$$

$$= 1 \oplus 1 \oplus 1$$

$$= 1$$

$$G_{13} = G_{12+1}(k) = 1 \oplus k_{(12 \bmod 10)+1} \oplus G_{12}(k)$$

$$= 1 \oplus 0 \oplus 1$$

$$= 0$$

$$G_{14} = G_{13+1}(k) = 1 \oplus k_{(13 \bmod 10)+1} \oplus G_{13}(k)$$

$$= 1 \oplus 0 \oplus 0$$

$$= 1$$

$$G_{15} = G_{14+1}(k) = 1 \oplus k_{(14 \bmod 10)+1} \oplus G_{14}(k)$$

$$= 1 \oplus 0 \oplus 1$$

$$= 0$$

$$G_{16} = G_{15+1}(k) = 1 \oplus k_{(15 \bmod 10)+1} \oplus G_{15}(k)$$

$$= 1 \oplus 1 \oplus 0$$

$$= 0$$

$$G_{17} = G_{16+1}(k) = 1 \oplus k_{(16 \bmod 10)+1} \oplus G_{16}(k)$$

$$= 1 \oplus 1 \oplus 0$$

$$= 0$$

$$G_{18} = G_{17+1}(k) = 1 \oplus k_{(17 \bmod 10)+1} \oplus G_{17}(k)$$

$$= 1 \oplus 1 \oplus 0$$

$$= 0$$

$$G_{19} = G_{18+1}(k) = 1 \oplus k_{(18 \bmod 10)+1} \oplus G_{18}(k)$$

$$= 1 \oplus 1 \oplus 0$$

$$= 0$$

$$G_{20} = G_{19+1}(k) = 1 \oplus k_{(19 \bmod 10)+1} \oplus G_{19}(k)$$

$$= 1 \oplus 1 \oplus 0$$

$$= 0$$

$$G_{21} = G_{20+1}(k) = 1 \oplus k_{(20 \bmod 10)+1} \oplus G_{20}(k)$$

$$= 1 \oplus 1 \oplus 0 = 0$$

$$\begin{aligned}
 G_{22} &= G_{21+1}(k) = 1 \oplus k_{(21 \bmod 10)+1} \oplus G_{21}(k) \\
 &= 1 \oplus 1 \oplus 0 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 G_{23} &= G_{22+1}(k) = 1 \oplus k_{(22 \bmod 10)+1} \oplus G_{22}(k) \\
 &= 1 \oplus 0 \oplus 0 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 G_{24} &= G_{23+1}(k) = 1 \oplus k_{(23 \bmod 10)+1} \oplus G_{23}(k) \\
 &= 1 \oplus 0 \oplus 1 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 G_{25} &= G_{24+1}(k) = 1 \oplus k_{(24 \bmod 10)+1} \oplus G_{24}(k) \\
 &= 1 \oplus 0 \oplus 0 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 G_{26} &= G_{25+1}(k) = 1 \oplus k_{(25 \bmod 10)+1} \oplus G_{25}(k) \\
 &= 1 \oplus 1 \oplus 1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 G_{27} &= G_{26+1}(k) = 1 \oplus k_{(26 \bmod 10)+1} \oplus G_{26}(k) \\
 &= 1 \oplus 1 \oplus 1 = 1
 \end{aligned}$$

$$\begin{aligned}
 G_{28} &= G_{27+1}(k) = 1 \oplus k_{(27 \bmod 10)+1} \oplus G_{27}(k) \\
 &= 1 \oplus 1 \oplus 1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 G_{29} &= G_{28+1}(k) = 1 \oplus k_{(28 \bmod 10)+1} \oplus G_{28}(k) \\
 &= 1 \oplus 1 \oplus 1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 G_{30} &= G_{29+1}(k) = 1 \oplus k_{(29 \bmod 10)+1} \oplus G_{29}(k) \\
 &= 1 \oplus 1 \oplus 1 \\
 &= 1
 \end{aligned}$$

$$G(k)_{1, \dots, 30} = 11000111111010000000010111111$$

$$\bullet m = 00111001101001010000$$

$$G(k) = 11000111111010000000$$

$$\begin{aligned}
 \text{Encrypt} &= m \oplus G(k) \\
 &= 1111110010100010000
 \end{aligned}$$

$$\begin{aligned}
 \bullet G_{11}(k) &= 1 \oplus k_1 \oplus G_{10}(k) \\
 &= 1 \oplus G_1(k) \oplus G_{10}(k) \\
 &= A(G(k))_{i=1, \dots, 10}
 \end{aligned}$$

$$\text{Therefore, } \Pr(A(G(k))_{i=1, \dots, 10} = G_{11}(k)) = 1$$

$$\text{Adv}(A) = \left| \Pr(A(G(k))_{i=1, \dots, 10}) = G_{\text{II}}(k) \right| - \frac{1}{2}$$

$$= \left| 1 - \frac{1}{2} \right| = \frac{1}{2}$$

• G is not a secure PRG so the stream cipher is not semantically secure.

3.

1. There are $4! = 24$ permutations
2. There are $4^4 = 256$ functions.
3. For $f(0)$, there are 4 possible values
 For $f(1)$, there are 3 possible values
 For $f(2)$, there are 4 possible values
 For $f(3)$, there are 4 possible values

Total there are $4 \times 3 \times 4 \times 4 = 192$ functions.

4. In a permutation, the probability that two distinct elements are mapped to different values is always 1, because permutations are one-to-one functions. Therefore, the adversary will always return 0.

$$\Pr(W_0) = 1$$

Experiment 1: $\Pr(w_1)$

In Experiment 1, the challenger picks a random function from the set of all functions $X \rightarrow X$, there are 256 possible functions. The probability that a randomly chosen function maps two distinct inputs to different value is

$$P(f(0) \neq f(1)) = \frac{192}{256} = \frac{3}{4} = \Pr(w_1)$$

$$\begin{aligned} \text{Advantage} &= |\Pr(w_0) - \Pr(w_1)| \\ &= \left| 1 - \frac{3}{4} \right| = \frac{1}{4} \text{ ans} \end{aligned}$$

$$4. \sqrt{2 \ln \left(\frac{1}{1-0.1} \right)} \cdot \sqrt{2^{224} + 1}$$

$$\sqrt{2 \ln \left(\frac{10}{9} \right)} \cdot 2^{112} + 1$$

$$2.383490668 \times 10^{33} + 1$$

$$\approx 2.3835 \times 10^{33} \text{ ans}$$